

Convenience Yield as the Shadow Price of Physical Security

Inventory, Systemic Risk, and Strategic Reserves

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Abstract

Convenience yield is usually defined as the implicit non-monetary benefit from physically holding a commodity rather than holding a claim to future delivery. This paper develops a stronger interpretation: convenience yield is the shadow price of physical security. When future delivery is uncertain, inventory provides insurance against stockouts, sanctions, blockades, logistical failure, war, and systemic financial disruption. We derive convenience yield as the marginal avoided disruption loss from physically available inventory. This derivation implies that convenience yield rises with systemic risk and falls with inventory abundance, not by assumption but by comparative statics. The model is embedded into a no-arbitrage futures-pricing framework with stochastic convenience yield, risk-neutral dynamics, and an affine term-structure approximation. We then extend the theory to strategic reserves, geoeconomic statecraft, warfare logistics, and international political economy. The empirical section proposes panel regressions, event studies, threshold tests, and machine-learning forecasting designs for estimating the convenience yield of security. The paper concludes that convenience yield should be understood not merely as a commodity-market residual, but as a measurable price of physical command over scarce economic resources under uncertainty.

The paper ends with “The End”

Keywords: convenience yield, inventories, futures pricing, systemic risk, strategic reserves, geopolitical risk, warfare economics, sanctions, commodity finance, machine learning.

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1 Introduction

Commodity finance traditionally treats convenience yield as the implicit benefit from holding physical inventory. A firm that possesses crude oil, natural gas, wheat, uranium, copper, or rare earth elements obtains more than the asset’s resale value. It obtains productive continuity, delivery certainty, option value, and resilience against future scarcity. These benefits become especially important during crises: wars, sanctions, blockades, pandemics, shipping interruptions, cyberattacks, export bans, financial freezes, and strategic embargoes.

This paper argues that convenience yield should be interpreted as the *shadow price of physical security*. Physical inventory is not merely a buffer stock. It is a security asset. It protects the holder against the failure of future markets, contracts, logistics, payments, and geopolitical access. When delivery promises become less credible, the value of possession rises.

The classical cost-of-carry formula defines the futures price of a storable commodity as

$$F_{t,T} = S_t \exp\{(r_t + u_t - y_{t,T})(T - t)\}, \quad (1)$$

where $F_{t,T}$ is the futures price, S_t is the spot price, r_t is the financing rate, u_t is the storage and insurance cost, and $y_{t,T}$ is the convenience yield. Rearranging gives the implied convenience yield

$$y_{t,T}^{\text{imp}} = r_t + u_t - \frac{1}{T-t} \log \left(\frac{F_{t,T}}{S_t} \right). \quad (2)$$

This formula identifies convenience yield as the residual that reconciles spot and futures prices. But it does not explain why the residual exists.

The central contribution of this paper is to derive convenience yield from a structural avoided-loss model. Let I_t denote inventory and let $\mathcal{R}_t$ denote systemic disruption risk. If the expected disruption loss from insufficient inventory is

$$L(I_t, \mathcal{R}_t) = \frac{\chi \mathcal{R}_t}{(I_t + \varepsilon)^\beta}, \quad \chi > 0, \quad \beta > 0, \quad \varepsilon > 0, \quad (3)$$

then the convenience yield is the marginal avoided loss from an additional unit of inventory:

$$y_t^{\text{str}} = -\frac{\partial L(I_t, \mathcal{R}_t)}{\partial I_t} = \frac{\chi \beta \mathcal{R}_t}{(I_t + \varepsilon)^{\beta+1}}. \quad (4)$$

This expression yields the desired comparative statics directly:

$$\frac{\partial y_t^{\text{str}}}{\partial \mathcal{R}_t} > 0, \quad \frac{\partial y_t^{\text{str}}}{\partial I_t} < 0. \quad (5)$$

Convenience yield rises with systemic risk and falls with inventory abundance because it is the marginal value of avoided disruption.

This paper makes six contributions. First, it interprets convenience yield as the shadow price of physical security under delivery uncertainty. Second, it derives the relation between convenience yield, inventories, and systemic risk from a loss function rather than assuming it. Third, it embeds the resulting yield into a no-arbitrage futures-pricing framework. Fourth, it develops a theory of strategic reserves in which optimal reserves increase with systemic risk. Fifth, it connects commodity inventories to geoeconomic statecraft, sanctions, international relations, and warfare logistics. Sixth, it proposes econometric and machine-learning methods for estimating and forecasting convenience yield under crisis conditions.

2 Classical Convenience Yield

2.1 Cost of Carry

Let S_t be the spot price of a commodity at time t and $F_{t,T}$ the futures price for delivery at time $T > t$. In a frictionless market with storage cost u_t , financing rate r_t , and convenience yield $y_{t,T}$, the cost-of-carry relation is

$$F_{t,T} = S_t \exp \{ (r_t + u_t - y_{t,T})(T - t) \}. \quad (6)$$

When $y_{t,T}$ is high, the futures price is low relative to the spot price. If

$$S_t > F_{t,T}, \quad (7)$$

the market is in backwardation. Backwardation is therefore consistent with a high convenience yield, although backwardation can also reflect risk premia, liquidity frictions, or segmented market participation.

Definition 1 (Implied convenience yield). *For a commodity with spot price S_t , futures price $F_{t,T}$, financing rate r_t , and storage cost u_t , the implied convenience yield is*

$$y_{t,T}^{\text{imp}} = r_t + u_t - \frac{1}{T-t} \log \left(\frac{F_{t,T}}{S_t} \right). \quad (8)$$

The implied convenience yield is observable from market data once storage and financing costs are specified. However, it is a reduced-form object. It combines physical scarcity, liquidity premia, expected delivery failures, basis risk, and measurement error.

2.2 Implied and Structural Convenience Yield

We distinguish between implied and structural convenience yield.

Definition 2 (Structural convenience yield). *The structural convenience yield is the marginal economic value of one additional unit of physically available inventory, holding spot price fixed.*

The implied yield is inferred from prices:

$$y_{t,T}^{\text{imp}} = r_t + u_t - \frac{1}{T-t} \log(F_{t,T}/S_t). \quad (9)$$

The structural yield is generated by inventory scarcity, systemic risk, and delivery fragility:

$$y_t^{\text{str}} = g(I_t, \mathcal{R}_t, \mathcal{D}_t), \quad (10)$$

where I_t is inventory, $\mathcal{R}_t$ is systemic risk, and $\mathcal{D}_t$ is delivery fragility.

A useful bridge equation is

$$y_{t,T}^{\text{imp}} = y_t^{\text{str}} + \pi_t^{\text{liq}} + \pi_t^{\text{del}} + \pi_{t,T}^{\text{basis}} + \nu_{t,T}, \quad (11)$$

where π_t^{liq} is a liquidity premium, π_t^{del} is a delivery-failure premium, $\pi_{t,T}^{\text{basis}}$ is a maturity-specific basis component, and $\nu_{t,T}$ is measurement error.

3 Inventory as Physical Security

3.1 Disruption Loss

Let $I_t \geq 0$ denote physical inventory. Let $\mathcal{R}_t \geq 0$ denote systemic disruption risk. This risk summarizes macroeconomic instability, war risk, geopolitical fragmentation, sanctions, shipping disruption, sovereign stress, financial contagion, energy-system vulnerability, and institutional breakdown.

Assumption 1 (Disruption loss). *Expected disruption loss is*

$$L(I_t, \mathcal{R}_t) = \frac{\chi \mathcal{R}_t}{(I_t + \varepsilon)^\beta}, \quad \chi > 0, \quad \beta > 0, \quad \varepsilon > 0. \quad (12)$$

The parameter χ measures exposure to disruption, β measures scarcity convexity, and ε prevents singularity at zero inventory. The loss function is decreasing and convex in inventory:

$$\frac{\partial L}{\partial I_t} = -\frac{\chi \beta \mathcal{R}_t}{(I_t + \varepsilon)^{\beta+1}} < 0, \quad (13)$$

and

$$\frac{\partial^2 L}{\partial I_t^2} = \frac{\chi \beta (\beta + 1) \mathcal{R}_t}{(I_t + \varepsilon)^{\beta+2}} > 0. \quad (14)$$

Definition 3 (Convenience yield as shadow price). *The structural convenience yield is the marginal avoided disruption loss from one additional unit of physical inventory:*

$$y_t^{\text{str}} = -\frac{\partial L(I_t, \mathcal{R}_t)}{\partial I_t}. \quad (15)$$

Therefore,

$$y_t^{\text{str}} = \frac{\chi \beta \mathcal{R}_t}{(I_t + \varepsilon)^{\beta+1}}. \quad (16)$$

Theorem 1 (Shadow-price theorem). *If expected disruption loss is*

$$L(I_t, \mathcal{R}_t) = \frac{\chi \mathcal{R}_t}{(I_t + \varepsilon)^\beta},$$

and convenience yield is the marginal avoided disruption loss, then convenience yield is increasing in systemic risk and decreasing in inventory abundance:

$$\frac{\partial y_t^{\text{str}}}{\partial \mathcal{R}_t} > 0, \quad \frac{\partial y_t^{\text{str}}}{\partial I_t} < 0.$$

Proof. By definition,

$$y_t^{\text{str}} = -\frac{\partial L}{\partial I_t} = \frac{\chi \beta \mathcal{R}_t}{(I_t + \varepsilon)^{\beta+1}}.$$

Differentiating with respect to $\mathcal{R}_t$ gives

$$\frac{\partial y_t^{\text{str}}}{\partial \mathcal{R}_t} = \frac{\chi \beta}{(I_t + \varepsilon)^{\beta+1}} > 0.$$

Differentiating with respect to I_t gives

$$\frac{\partial y_t^{\text{str}}}{\partial I_t} = -\frac{\chi \beta (\beta + 1) \mathcal{R}_t}{(I_t + \varepsilon)^{\beta+2}} < 0.$$

Thus convenience yield rises with systemic risk and falls with inventories. $\square$

3.2 Risk Amplification at Low Inventories

The cross-partial derivative is

$$\frac{\partial^2 y_t^{\text{str}}}{\partial I_t \partial \mathcal{R}_t} = -\frac{\chi\beta(\beta+1)}{(I_t + \varepsilon)^{\beta+2}} < 0. \quad (17)$$

This means systemic risk has a larger effect on convenience yield when inventories are low. Scarcity amplifies risk.

Corollary 1 (Scarcity amplification). *For any $\Delta\mathcal{R} > 0$, the increase in convenience yield caused by an increase in systemic risk is larger at lower inventory levels.*

Proof. The marginal effect of systemic risk is

$$\frac{\partial y_t^{\text{str}}}{\partial \mathcal{R}_t} = \frac{\chi\beta}{(I_t + \varepsilon)^{\beta+1}},$$

which is strictly decreasing in I_t . □

4 A Systemic Risk Decomposition

Systemic disruption risk is multidimensional. We write

$$\mathcal{R}_t = w_1 \mathcal{M}_t + w_2 \mathcal{G}_t + w_3 \mathcal{F}_t + w_4 \mathcal{W}_t + w_5 \mathcal{L}_t, \quad (18)$$

where $\mathcal{M}_t$ is macroeconomic risk, $\mathcal{G}_t$ geopolitical risk, $\mathcal{F}_t$ financial stress, $\mathcal{W}_t$ warfare risk, and $\mathcal{L}_t$ logistical fragility. We assume $w_j \geq 0$ and $\sum_{j=1}^5 w_j = 1$.

Table 1: Components of systemic disruption risk

Component	Examples	Expected effect on convenience yield
Macroeconomic risk	Inflation uncertainty, recession risk, currency stress	Raises precautionary inventory value
Geopolitical risk	Sanctions, export controls, alliance fragmentation	Raises delivery-failure premium
Financial stress	Credit freeze, margin spiral, collateral shortage	Raises value of physical possession over promises
Warfare risk	Blockade, bombardment, mobilization, military demand	Raises operational-continuity premium
Logistical fragility	Port closure, shipping disruption, pipeline sabotage	Raises short-term scarcity premium

The structural yield becomes

$$y_t^{\text{str}} = \frac{\chi\beta}{(I_t + \varepsilon)^{\beta+1}} (w_1 \mathcal{M}_t + w_2 \mathcal{G}_t + w_3 \mathcal{F}_t + w_4 \mathcal{W}_t + w_5 \mathcal{L}_t). \quad (19)$$

5 Futures Pricing with Stochastic Convenience Yield

5.1 Risk-Neutral Dynamics

Under the risk-neutral measure $\mathbb{Q}$, let the spot price satisfy

$$\frac{dS_t}{S_t} = (r_t + u_t - y_t) dt + \sigma_S dW_t^{\mathbb{Q}}. \quad (20)$$

Let the convenience yield follow a mean-reverting process:

$$dy_t = \kappa(\theta - y_t) dt + \eta dZ_t^{\mathbb{Q}}, \quad (21)$$

with

$$dW_t^{\mathbb{Q}} dZ_t^{\mathbb{Q}} = \rho dt. \quad (22)$$

Here $\kappa > 0$ is the speed of mean reversion, θ is the long-run yield, η is convenience-yield volatility, and ρ is the correlation between spot-price shocks and convenience-yield shocks.

Proposition 1 (No-arbitrage futures price). *If the spot price follows*

$$\frac{dS_t}{S_t} = (r_t + u_t - y_t) dt + \sigma_S dW_t^{\mathbb{Q}},$$

then the futures price satisfies

$$F(t, T) = \mathbb{E}_t^{\mathbb{Q}}[S_T].$$

Proof. A futures contract has zero initial cost and daily settlement. Under no arbitrage, the futures price equals the risk-neutral conditional expectation of the terminal spot price when interest rates are deterministic or when futures and forward convexity adjustments are ignored. Therefore,

$$F(t, T) = \mathbb{E}_t^{\mathbb{Q}}[S_T].$$

□

5.2 Affine Approximation

When $r_t = r$, $u_t = u$, and y_t follows an Ornstein–Uhlenbeck process under $\mathbb{Q}$, the futures price admits the affine approximation

$$F(t, T) = S_t \exp\{A(\tau) - B(\tau)y_t\}, \quad \tau = T - t, \quad (23)$$

where

$$B(\tau) = \frac{1 - e^{-\kappa\tau}}{\kappa}. \quad (24)$$

The function $A(\tau)$ collects financing, storage, long-run convenience yield, volatility, and convexity terms.

Lemma 1 (Term-structure sensitivity). *The sensitivity of the log futures price to current convenience yield is*

$$\frac{\partial \log F(t, T)}{\partial y_t} = -B(\tau) = -\frac{1 - e^{-\kappa\tau}}{\kappa} < 0.$$

Proof. Taking logs,

$$\log F(t, T) = \log S_t + A(\tau) - B(\tau)y_t.$$

Differentiating with respect to y_t gives

$$\frac{\partial \log F(t, T)}{\partial y_t} = -B(\tau).$$

Since $\kappa > 0$ and $\tau > 0$, $B(\tau) > 0$, so the derivative is negative. □

The lemma formalizes a standard intuition: a higher convenience yield lowers futures prices relative to spot prices.

6 Strategic Reserves

6.1 Optimal Inventory

Consider a state, firm, or military planner that chooses inventory $I \geq 0$ to minimize storage cost plus expected disruption loss:

$$\min_{I \geq 0} cI + \frac{\chi \mathcal{R}}{(I + \varepsilon)^\beta}, \quad (25)$$

where $c > 0$ is the marginal storage cost.

The first-order condition is

$$c = \frac{\chi \beta \mathcal{R}}{(I^* + \varepsilon)^{\beta+1}}. \quad (26)$$

Solving,

$$I^* = \left(\frac{\chi \beta \mathcal{R}}{c} \right)^{1/(\beta+1)} - \varepsilon. \quad (27)$$

Theorem 2 (Strategic reserve theorem). *Suppose the interior solution is positive. Then optimal inventory satisfies*

$$\frac{\partial I^*}{\partial \mathcal{R}} > 0, \quad \frac{\partial I^*}{\partial c} < 0.$$

Thus strategic reserves increase with systemic disruption risk and decrease with storage cost.

Proof. From

$$I^* = \left(\frac{\chi^\beta \mathcal{R}}{c} \right)^{1/(\beta+1)} - \varepsilon,$$

differentiate with respect to $\mathcal{R}$:

$$\frac{\partial I^*}{\partial \mathcal{R}} = \frac{1}{\beta+1} \left(\frac{\chi^\beta}{c} \right)^{1/(\beta+1)} \mathcal{R}^{-\beta/(\beta+1)} > 0.$$

Differentiating with respect to c gives

$$\frac{\partial I^*}{\partial c} = -\frac{1}{\beta+1} (\chi^\beta \mathcal{R})^{1/(\beta+1)} c^{-(\beta+2)/(\beta+1)} < 0.$$

□

6.2 Strategic Reserve Ratio

Define the strategic reserve ratio as

$$\text{SRR}_t = \frac{I_t}{D_t}, \quad (28)$$

where D_t is expected consumption demand over a security horizon. A country with higher SRR_t has a lower probability of operational stockout.

Let the probability of stockout be

$$p_t = \exp(-\lambda \text{SRR}_t), \quad \lambda > 0. \quad (29)$$

Then the expected disruption loss can be written as

$$L_t = \Omega_t p_t = \Omega_t \exp(-\lambda I_t / D_t), \quad (30)$$

where Ω_t is the cost of failure. The associated convenience yield is

$$y_t = -\frac{\partial L_t}{\partial I_t} = \frac{\lambda \Omega_t}{D_t} \exp(-\lambda I_t / D_t). \quad (31)$$

This alternative specification implies the same qualitative logic: physical inventories have high marginal value when inventories are low, demand is high, and failure costs are severe.

7 Warfare Economics

7.1 Military Production and Energy Security

Let military output be

$$Q_t = A_t K_t^\alpha E_t^\beta L_t^\gamma, \quad \alpha, \beta, \gamma > 0, \quad (32)$$

where K_t is capital, E_t is energy availability, L_t is mobilized labor, and A_t is organizational effectiveness.

If energy availability is constrained by inventory,

$$E_t = \min\{\bar{E}_t, I_t + \Delta_t\}, \quad (33)$$

where $\bar{E}_t$ is desired energy use and Δ_t is uncertain delivery, then the marginal military value of inventory is

$$\frac{\partial Q_t}{\partial I_t} = A_t K_t^\alpha L_t^\gamma \beta E_t^{\beta-1} \frac{\partial E_t}{\partial I_t}. \quad (34)$$

When delivery is unreliable and $E_t = I_t + \Delta_t < \bar{E}_t$, we have $\partial E_t / \partial I_t = 1$, and therefore

$$\frac{\partial Q_t}{\partial I_t} = A_t K_t^\alpha L_t^\gamma \beta E_t^{\beta-1} > 0. \quad (35)$$

Proposition 2 (Operational-continuity premium). *In a war economy where energy availability constrains military output, the convenience yield of energy inventory includes an operational-continuity premium proportional to the marginal military product of energy inventory.*

Proof. If energy inventory relaxes the constraint on military output, then one additional unit of inventory raises output by $\partial Q_t / \partial I_t > 0$. The planner values this increment at the shadow value of output, say $\Lambda_t > 0$. Hence the operational-continuity premium is

$$\Lambda_t \frac{\partial Q_t}{\partial I_t} > 0.$$

This premium enters the total convenience yield of energy inventories. □

7.2 Warfare Channels

Table 2: Warfare channels raising convenience yield

Channel	Mechanism	Convenience-yield effect
Mobilization demand	Sudden increase in fuel, metals, food, and logistics demand	Higher marginal value of inventory
Blockade risk	Maritime or land-route disruption	Higher delivery-failure premium
Sanctions	Legal and financial exclusion from supply networks	Higher possession value
Infrastructure attack	Pipeline, refinery, port, grid, or rail damage	Higher operational-continuity premium
Alliance fragmentation	Loss of trusted suppliers	Higher strategic-reserve demand

8 Geopolitics and International Political Economy

Commodity inventories are embedded in international power structures. Control over strategic commodities can create bargaining power, coercive leverage, and vulnerability.

Let country j have strategic commodity security

$$\Sigma_{j,t} = \omega_1 I_{j,t} + \omega_2 P_{j,t} + \omega_3 A_{j,t} - \omega_4 V_{j,t}, \quad (36)$$

where $I_{j,t}$ is inventory, $P_{j,t}$ domestic production capacity, $A_{j,t}$ alliance reliability, and $V_{j,t}$ import vulnerability. A country with high $\Sigma_{j,t}$ faces a lower convenience yield because future availability is more secure.

A simple geopolitical convenience-yield equation is

$$y_{j,t}^{\text{geo}} = \frac{\phi \mathcal{G}_t}{\Sigma_{j,t} + \varepsilon}, \quad \phi > 0. \quad (37)$$

Thus geopolitical convenience yield is high when geopolitical risk is high and national supply security is low.

Table 3: Political-economy interpretation of physical inventory

Discipline	Interpretation of inventory	Convenience-yield meaning
Economics	Buffer against scarcity and price spikes	Marginal avoided stockout loss
Finance	State variable in futures pricing	Implied yield in cost of carry
Political science	Material basis of policy autonomy	Sovereignty premium
Geopolitics	Control over strategic resources	Coercion-resistance premium
International relations	Dependence within inter-state networks	Alliance and supply-chain risk premium
Warfare economics	Input into mobilization and logistics	Operational-continuity premium

9 Vector Graphics

9.1 Convenience Yield Mechanism

Convenience yield as shadow price of physical security

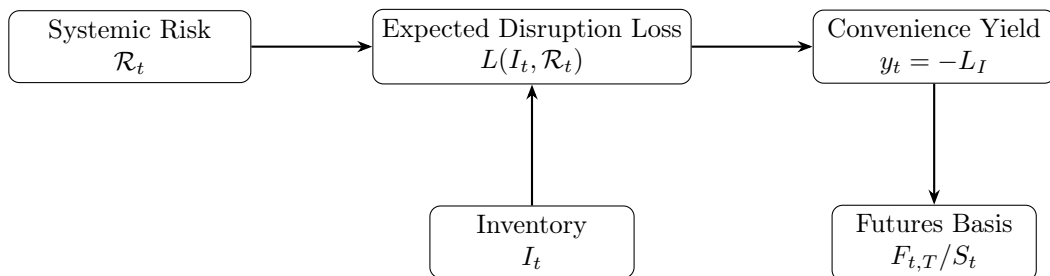


Figure 1: The structural channel from systemic risk and inventories to convenience yield.

9.2 Inventory Scarcity and Convenience Yield

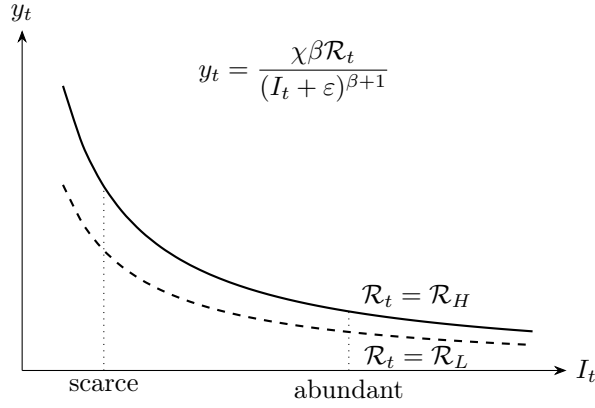


Figure 2: Convenience yield is high when inventories are scarce and systemic risk is elevated.

10 Empirical Identification

10.1 Constructing Implied Convenience Yield

For commodity i , maturity T , and time t , construct

$$y_{i,t,T}^{\text{imp}} = r_t + u_{i,t} - \frac{1}{T-t} \log \left(\frac{F_{i,t,T}}{S_{i,t}} \right). \quad (38)$$

Storage cost $u_{i,t}$ can be proxied by warehouse cost, insurance, financing spreads, freight rates, or estimated as a commodity-specific latent factor.

10.2 Panel Regression

A baseline panel specification is

$$y_{i,t,T}^{\text{imp}} = \alpha_i + \lambda_T + \delta_t + \beta_1 \mathcal{R}_t + \beta_2 I_{i,t}^{-1} + \beta_3 (\mathcal{R}_t \times I_{i,t}^{-1}) + \Gamma X_{i,t} + \varepsilon_{i,t,T}. \quad (39)$$

Here α_i are commodity fixed effects, λ_T are maturity fixed effects, δ_t are time fixed effects, and $X_{i,t}$ includes volatility, freight cost, interest rates, seasonality, open interest, and liquidity measures.

The central hypothesis is

$$\beta_3 > 0. \quad (40)$$

This means systemic risk has a stronger effect on convenience yield when inventories are low.

10.3 Event Study

Let e index geopolitical events such as sanctions, export bans, wars, blockades, pipeline disruptions, shipping chokepoint closures, or financial freezes. Let t_e denote the event date. Estimate

$$\Delta y_{i,t,T}^{\text{imp}} = \alpha_i + \lambda_T + \sum_{k=-K}^K \beta_k \mathbf{1}\{t - t_e = k\} + \Gamma X_{i,t} + \varepsilon_{i,t,T}. \quad (41)$$

The event-study prediction is

$$\beta_k > 0 \quad \text{for post-event windows involving strategic commodities.} \quad (42)$$

10.4 Threshold Model

Risk effects may become nonlinear when inventories cross a scarcity threshold $\bar{I}_i$. Define

$$\text{Scarce}_{i,t} = \mathbf{1}\{I_{i,t} < \bar{I}_i\}. \quad (43)$$

Then estimate

$$y_{i,t,T}^{\text{imp}} = \alpha_i + \lambda_T + \delta_t + \beta_1 \mathcal{R}_t + \beta_2 \text{Scarce}_{i,t} + \beta_3 (\mathcal{R}_t \times \text{Scarce}_{i,t}) + \Gamma X_{i,t} + \varepsilon_{i,t,T}. \quad (44)$$

The predicted sign is

$$\beta_3 > 0. \quad (45)$$

Table 4: Empirical predictions

Variable	Symbol	Expected sign	Interpretation
Systemic risk	$\mathcal{R}_t$	+	Risk raises value of possession
Inverse inventory	$I_{i,t}^{-1}$	+	Scarcity raises convenience yield
Risk–scarcity interaction	$\mathcal{R}_t \times I_{i,t}^{-1}$	+	Risk matters more when inventories are low
Storage cost	$u_{i,t}$	ambiguous	May proxy scarcity or carrying cost
Liquidity	$\text{Liq}_{i,t}$	–	More liquid markets reduce possession premium
Delivery fragility	$\mathcal{D}_{i,t}$	+	Fragile logistics raise delivery-failure premium

11 Statistical Framework

11.1 State-Space Model

Convenience yield can be treated as a latent state. Let the measurement equation be

$$y_{i,t,T}^{\text{imp}} = y_{i,t}^{\text{str}} + b_{i,T} + \nu_{i,t,T}, \quad \nu_{i,t,T} \sim N(0, \sigma_\nu^2). \quad (46)$$

Let the transition equation be

$$y_{i,t}^{\text{str}} = \mu_i + \rho_i y_{i,t-1}^{\text{str}} + \theta_1 \mathcal{R}_t + \theta_2 I_{i,t}^{-1} + \theta_3 \mathcal{R}_t I_{i,t}^{-1} + \zeta_{i,t}, \quad (47)$$

with

$$\zeta_{i,t} \sim N(0, \sigma_\zeta^2). \quad (48)$$

This model separates observed futures-basis variation from the latent structural convenience yield.

11.2 Hypothesis Testing

The main null hypothesis is

$$H_0 : \theta_3 = 0, \quad (49)$$

against

$$H_1 : \theta_3 > 0. \quad (50)$$

Rejection of H_0 supports the scarcity-amplification mechanism.

11.3 Robustness Checks

Robustness checks should include:

1. alternative storage-cost proxies;
2. alternative systemic-risk indices;
3. exclusion of crisis periods;
4. crisis-only subsamples;
5. commodity-by-commodity estimation;
6. maturity-specific futures curves;
7. instrumental-variable specifications using exogenous logistics shocks;
8. placebo tests on commodities without strategic scarcity;
9. pre-trend tests in event studies;
10. out-of-sample forecasting evaluation.

12 Algorithms

Table 5: Algorithm 1: Construction of implied convenience yield

Step	Operation
1	Collect spot prices $S_{i,t}$ and futures prices $F_{i,t,T}$ for commodity i and maturity T .
2	Estimate or proxy financing rate r_t and storage cost $u_{i,t}$.
3	Compute $y_{i,t,T}^{\text{imp}} = r_t + u_{i,t} - (T - t)^{-1} \log(F_{i,t,T}/S_{i,t})$.
4	Winsorize extreme observations caused by illiquid contracts or bad quotes.
5	Merge with inventory, volatility, freight, macroeconomic, and geopolitical variables.
6	Estimate panel, state-space, threshold, and event-study models.

Table 6: Algorithm 2: Walk-forward machine-learning forecast

Step	Operation
1	Choose forecast horizon h and target $y_{i,t+h,T}^{\text{imp}}$.
2	Construct feature vector $X_{i,t}$ using inventories, futures slopes, volatility, freight costs, risk indices, and event indicators.
3	Train models on data up to time t .
4	Forecast $y_{i,t+h,T}^{\text{imp}}$.
5	Move the training window forward and repeat.
6	Compare models using mean squared error, mean absolute error, directional accuracy, and crisis-period loss.

Table 7: Algorithm 3: Strategic-reserve stress test

Step	Operation
1	Specify a crisis scenario: war, sanctions, blockade, cyberattack, pandemic, or export ban.
2	Shock systemic risk $\mathcal{R}_t$, delivery fragility $\mathcal{D}_t$, and expected demand D_t .
3	Compute structural yield $y_t^{\text{str}} = \chi \beta \mathcal{R}_t / (I_t + \varepsilon)^{\beta+1}$.
4	Solve for optimal reserve $I^* = [\chi \beta \mathcal{R}_t / c]^{1/(\beta+1)} - \varepsilon$.
5	Compare actual inventory I_t with optimal inventory I^* .
6	Report reserve gap $G_t = I^* - I_t$ and implied vulnerability.

13 Machine Learning and AI

13.1 Prediction Target

The machine-learning target is the future implied convenience yield:

$$Y_{i,t+h,T} = y_{i,t+h,T}^{\text{imp}} \quad (51)$$

The feature vector is

$$X_{i,t} = [I_{i,t}^{-1}, \mathcal{R}_t, \mathcal{G}_t, \mathcal{F}_t, \mathcal{W}_t, \sigma_{i,t}, \text{Freight}_t, \text{Spread}_{i,t}, \text{Slope}_{i,t,T}, \text{Season}_{i,t}]. \quad (52)$$

A supervised learner estimates

$$\hat{y}_{i,t+h,T} = f_{\theta}(X_{i,t}). \quad (53)$$

The loss function is

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i,t,T} \left(y_{i,t+h,T}^{\text{imp}} - f_{\theta}(X_{i,t}) \right)^2. \quad (54)$$

13.2 Model Classes

Table 8: Machine-learning models for convenience-yield forecasting

Model	Strength	Role in the paper
AR(1) and ARIMA	Captures persistence	Baseline benchmark
LASSO and elastic net	Sparse selection of macro and risk variables	Interpretable benchmark
Random forest	Nonlinearities and interactions	Detects inventory-risk thresholds
Gradient boosting	Strong tabular prediction	Main forecasting benchmark
Neural network	Flexible nonlinear approximation	Crisis-regime forecasting
LSTM or temporal network	Sequential dependence	Futures-curve dynamics
Reinforcement learning	Sequential inventory policy	Strategic-reserve optimization

13.3 Representation Learning

Let $Z_{i,t} = f_\phi(X_{i,t})$ be a learned latent representation of commodity-market security conditions. A prediction head g_ψ maps $Z_{i,t}$ into the forecast:

$$\hat{y}_{i,t+h,T} = g_\psi(f_\phi(X_{i,t})). \quad (55)$$

The representation $Z_{i,t}$ can be interpreted as a latent security state combining inventories, logistics, volatility, geopolitical risk, and financial stress.

Definition 4 (Latent security state). *A latent security state is a low-dimensional representation $Z_{i,t}$ that is sufficient for forecasting future convenience yield conditional on the available feature set:*

$$\mathbb{E}[y_{i,t+h,T}^{\text{imp}} \mid X_{i,t}] = \mathbb{E}[y_{i,t+h,T}^{\text{imp}} \mid Z_{i,t}].$$

13.4 Reinforcement Learning for Reserve Policy

Let the state be

$$s_t = (I_t, \mathcal{R}_t, D_t, c_t, S_t, F_{t,T}), \quad (56)$$

and let the action a_t be the reserve adjustment:

$$I_{t+1} = I_t + a_t - \text{Use}_t. \quad (57)$$

A planner maximizes expected discounted reward

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \delta^t (-B(a_t) - c_t I_t - L(I_t, \mathcal{R}_t)) \right], \quad (58)$$

where $B(a_t)$ is the adjustment cost. The Bellman equation is

$$V(s_t) = \max_{a_t} \{-B(a_t) - c_t I_t - L(I_t, \mathcal{R}_t) + \delta \mathbb{E}[V(s_{t+1}) \mid s_t, a_t]\}. \quad (59)$$

This converts strategic reserve management into a Markov decision process.

14 Simulation Design

A simulation can be used to illustrate the theory. Let inventory follow

$$I_{t+1} = I_t + q_t - d_t + \xi_t, \quad (60)$$

where q_t is replenishment, d_t is demand, and ξ_t is a supply shock. Let systemic risk follow

$$\mathcal{R}_{t+1} = \rho_R \mathcal{R}_t + \sigma_R \epsilon_{t+1} + J_{t+1}, \quad (61)$$

where J_{t+1} is a jump representing war, sanctions, blockade, or financial crisis. Convenience yield is

$$y_t = \frac{\chi \beta \mathcal{R}_t}{(I_t + \varepsilon)^{\beta+1}}. \quad (62)$$

The simulated spot-price process is

$$\frac{dS_t}{S_t} = (\mu - y_t) dt + \sigma_S dW_t. \quad (63)$$

Table 9: Illustrative simulation regimes

Regime	Inventory	Risk	Yield	Interpretation
Normal abundance	High	Low	Low	Standard commodity market
Macroeconomic stress	Medium	Medium	Medium	Precautionary inventory demand
Geopolitical shock	Medium	High	High	Delivery uncertainty
War scarcity	Low	High	Very high	Operational continuity dominates
Strategic reserve buildup	Rising	High	Falling	Policy reduces scarcity premium

15 Policy Implications

The framework implies that strategic inventory policy is not merely a storage problem. It is a security problem. A government that ignores convenience yield may underinvest in reserves because it observes carrying costs but fails to price avoided disruption losses.

The optimal reserve rule

$$I^* = \left(\frac{\chi \beta \mathcal{R}}{c} \right)^{1/(\beta+1)} - \varepsilon \quad (64)$$

implies that reserves should rise during periods of elevated geopolitical fragmentation, war risk, sanctions risk, or supply-chain fragility. The same logic applies to firms with critical inputs. The private optimum may differ from the social optimum because individual firms do not internalize macroeconomic spillovers from production stoppages.

Let the social disruption loss be

$$L^S(I, \mathcal{R}) = \frac{\chi_S \mathcal{R}}{(I + \varepsilon)^\beta}, \quad (65)$$

and the private disruption loss be

$$L^P(I, \mathcal{R}) = \frac{\chi_P \mathcal{R}}{(I + \varepsilon)^\beta}. \quad (66)$$

If $\chi_S > \chi_P$, private actors underinvest in reserves:

$$I_S^* > I_P^*. \quad (67)$$

Proposition 3 (Reserve externality). *If social disruption exposure exceeds private disruption exposure, then decentralized reserve holdings are below the social optimum.*

Proof. The optimal reserve is increasing in χ :

$$I^*(\chi) = \left(\frac{\chi \beta \mathcal{R}}{c} \right)^{1/(\beta+1)} - \varepsilon.$$

If $\chi_S > \chi_P$, then $I^*(\chi_S) > I^*(\chi_P)$. □

16 Limitations

The model intentionally abstracts from several complications. First, storage costs may be nonlinear and capacity constrained. Second, futures markets may be segmented across maturities and delivery locations. Third, implied convenience yield may contain risk premia unrelated to physical inventory. Fourth, geopolitical risk is difficult to measure and may be endogenous to commodity prices. Fifth, strategic reserves may create moral hazard if private actors expect public stabilization. Sixth, machine-learning forecasts may perform well in normal periods but fail under unprecedented crises.

These limitations do not undermine the central theorem. They indicate where empirical work must be careful: convenience yield is observable as a futures-market residual, but its structural interpretation requires identification.

17 Conclusion

Convenience yield is usually treated as a commodity-market residual. This paper develops a stronger interpretation: convenience yield is the shadow price of physical security. When future delivery is reliable, the benefit of holding inventory is modest. When future delivery is threatened by scarcity, war, sanctions, financial stress, logistical failure, or geopolitical fragmentation, physical possession becomes valuable.

The key theoretical result is simple. If expected disruption loss is decreasing and convex in inventory, then the convenience yield equals the marginal avoided disruption loss. This makes convenience yield increasing in systemic risk and decreasing in inventory abundance. The result links commodity finance to macroeconomics, political economy, international relations, and warfare economics.

The practical implication is equally direct. Inventories are not merely working capital. For strategic commodities, they are security assets. Their convenience yield measures the value of command over physical resources when promises, contracts, and future delivery channels become uncertain.

A Derivation of the Structural Yield

Let

$$L(I, \mathcal{R}) = \chi \mathcal{R} (I + \varepsilon)^{-\beta}.$$

Then

$$\frac{\partial L}{\partial I} = -\chi \beta \mathcal{R} (I + \varepsilon)^{-\beta-1}.$$

Therefore

$$y = -L_I = \chi \beta \mathcal{R} (I + \varepsilon)^{-\beta-1}.$$

The second derivative of L is

$$L_{II} = \chi \beta (\beta + 1) \mathcal{R} (I + \varepsilon)^{-\beta-2} > 0.$$

Thus losses are convex in inventory scarcity.

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Glossary

Backwardation A futures-market state in which the spot price exceeds the futures price, often associated with high convenience yield or immediate scarcity.

Convenience yield The non-monetary benefit from physically holding a commodity rather than holding a claim to future delivery.

Delivery fragility The vulnerability of future delivery to logistics failures, sanctions, blockades, infrastructure attacks, or institutional disruption.

Geoeconomic statecraft The use of economic instruments such as sanctions, export controls, financial restrictions, and supply-chain leverage for strategic political purposes.

Implied convenience yield The convenience yield inferred from the cost-of-carry relationship between spot and futures prices.

Operational-continuity premium The component of convenience yield arising from the value of uninterrupted production, logistics, or military capability.

Scarcity amplification The property that systemic risk has a larger effect on convenience yield when inventories are low.

Shadow price The implicit marginal value of relaxing a constraint; here, the marginal value of one additional unit of physical inventory.

Strategic reserve Inventory held not merely for ordinary commercial use but for national security, macroeconomic stabilization, or crisis resilience.

Systemic disruption risk The composite risk that future delivery, production, trade, finance, or logistics will fail because of macroeconomic, geopolitical, military, or institutional shocks.

War economy An economic system in which production, logistics, inventories, labor, and finance are reorganized around military objectives and national survival.

The End